3.1.1 Machines and Matrices

States in quantum mechanics are mathematically described as vectors in a vector space. Physical observables—the things that you can measure—are described by linear operators. We'll take that as an axiom, and we'll find out later (in Section 3.1.5) that operators corresponding to physical observables must be Hermitian as well as linear. The correspondence between operators and observables is subtle, and understanding it will take some effort.

Observables are the things you measure. For example, we can make direct measurements of the coordinates of a particle; the energy, momentum, or angular momentum of a system; or the electric field at a point in space. Observables are also associated with a vector space, but they are not state-vectors. They are the things you measure—σ_x would be an example—and they are represented by *linear operators*. John Wheeler liked to call such mathematical objects *machines*. He imagined a machine with two ports: an input port and an output port. In the input port you insert a vector, such as $|A\rangle$. The gears turn and the machine delivers a result in the output port. This result is another vector, say $|B\rangle$.

Let's denote the operator by the boldface letter **M** (for "machine"). Here is the equation to express the fact that **M** acts on the vector $|A\rangle$ to give $|B\rangle$:

$$\mathbf{M}|A\rangle = |B\rangle.$$

Not every machine is a *linear* operator. Linearity implies a few simple properties. To begin with, a linear operator must give a unique output for every vector in the space. We can imagine a machine that gives an output for some vectors, but just grinds up others and gives nothing. This machine would not be a linear operator. Something must come out for anything you put in.

The next property states that when a linear operator **M** acts on a multiple of an input vector, it gives the same multiple of the output vector. Thus, if